

Data Project #1

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1.1 Team analysis

This project will focus on studying the special effect of the arrival of a big star, and the influence on demand and revenues in the North American team of MLS will be examined. To be specific, the anchor team will be the Inter Miami CF from MLS, between the years 2018 and 2024, treating the year 2020(the COVID year) as the robust year. The angle will be the Messi effect on the economic shock. Whether Messi's arrival on the team of Inter Miami in 2023 brings huge influence on not only the league but the whole MLS's revenue and audience participation will be tested. It is also tested whether smaller markets seem to benefit more than larger markets from the superstar effect, by using the theory of competitive balance themes. Is it (the effect) disproportionately?

Speaking of the market, the Inter Miami CF has a tourism-heavy market, where South Florida is always full of visitors from all around the world, and it is filled with plenty of entertainment choices and is considered a high-income market. That raises the price of attention from the world, thus making the base for a recognizable star power, to have a more significant marginal effect on people's willingness to pay for the ticket. For the capacity constraint, their stadium in Fort Lauderdale, Florida, limited the seats offered, which creates a short-run supply cap. For the team, this may make a potential soar of ticket prices, and even secondary market spreads if the demand is super high in some seasons, especially after the arrival of a superstar like Messi to the team. It could be adjusted to offer more seats and thus expand the team's revenue to a new level. Moreover, Messi's influence even beyond the geographical factors. By promoting social media, his brand extends to worldwide fans, so out-of-market analysis is also very important for our project. Moreover, since smaller markets rely more on visiting stars' demand, the shock brought by Messi will flatten the revenue impermanently, which is the idea of testing the competitive balance in the following section 1.2.

To conclude, the Inter Miami CF here can be seen as a clean case to test the big star effect under capacity constraints in a developing team.

1.2 League analysis

For this part, a data framework is used to test two hypotheses about the Messi effect on MLS: one is that these teams without Messi's demand and revenue shifting after July 2023, when Messi was introduced to the team of Inter Miami; the second is that the percentage gains are larger in small markets, which states an improving competitive balance.

Data are from 2018 to 2024 (including both pre-Messi time and post-Messi time). Team revenue is collected from Professor Rodney Fort's website, then adjusted for inflation using

2023 as a base year, to ensure the enough pooled observations for the regression analysis across years. The percentage of capacity is generated from Transfermarkt, which will be further used for capacity constraints. Performance (points/playoffs) is from Forbes' website, which is motivated to control for the points in the regression model. Data are cleaned and stored in Excel.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
	team_id	Team	Year	Revenue_real	PctCapacity	MarketSize	InterMiami	Post	InterMiami_Post	NonMiami_Post	Superstar	Superstar_MarketSize	Covid	Points	Playoff	
124	3	Chicago Fire	2023	25	0.295	9.7	0	1	0	1	0	0	0	40	0	
125	4	Colorado Rapids	2023	33	0.458	3.09	0	1	0	1	0	0	0	27	0	
126	5	Columbus Crew	2023	55	1.015	2.17	0	1	0	1	0	0	0	57	1	
127	6	D.C. United	2023	70	0.877	6.54	0	1	0	1	0	0	0	40	0	
128	7	FC Dallas	2023	40	0.89	8.11	0	1	0	1	0	0	0	46	1	
129	8	Houston Dynamo	2023	39	0.682	7.35	0	1	0	1	0	0	0	51	1	
130	9	Inter Miami CF CF	2023	56	0.809	6.33	1	1	1	0	0	0	0	43	1	
131	10	LA Galaxy	2023	98	0.893	13.2	0	1	0	1	0	0	0	36	0	
132	11	LAFC	2023	0	1.007	13.2	0	1	0	1	0	0	0	52	1	
133	12	Minnesota United	2023	54	0.998	3.74	0	1	0	1	0	0	0	41	0	
134	13	New England Revolution	2023	37	0.348	4.97	0	1	0	1	0	0	0	55	1	
135	14	New York City FC	2023	55	0.668	19.8	0	1	0	1	0	0	0	41	0	
136	15	New York Red Bulls	2023	50	0.73	19.8	0	1	0	1	0	0	0	43	1	
137	16	Orlando City SC	2023	42	0.807	2.77	0	1	0	1	0	0	0	63	1	
138	17	Philadelphia Union	2023	54	1.023	6.3	0	1	0	1	0	0	0	55	1	
139	18	Portland Timbers	2023	65	0.916	2.57	0	1	0	1	0	0	0	43	0	
140	19	Real Salt Lake	2023	43	0.961	1.3	0	1	0	1	0	0	0	50	1	
141	20	San Jose Earthquakes	2023	43	1.023	2.05	0	1	0	1	0	0	0	44	1	
142	21	Seattle Sounders	2023	66	0.468	4.07	0	1	0	1	0	0	0	53	1	
143	22	Sporting Kansas City	2023	59	0.887	2.25	0	1	0	1	0	0	0	44	1	
144	23	Toronto FC	2023	62	0.817	6.77	0	1	0	1	0	0	0	22	0	
145	24	Vancouver Whitecaps	2023	21	0.307	2.79	0	1	0	1	0	0	0	48	1	
146	1	Atlanta United	2024	130.8139535	0.676	6.41	0	1	0	1	0	0	0	50	1	
147	2	CF Montreal	2024	27.13179295	0.942	4.65	0	1	0	1	0	0	0	42	0	
148	3	Chicago Fire	2024	24.2248062	0.352	9.73	0	1	0	1	0	0	0	44	0	
149	4	Colorado Rapids	2024	53.29457364	0.451	3.11	0	1	0	1	0	0	0	43	0	
150	5	Columbus Crew	2024	58.13953488	1.032	2.18	0	1	0	1	0	0	0	61	1	
151	6	D.C. United	2024	87.20930233	0.903	6.57	0	1	0	1	0	0	0	40	0	
152	7	FC Dallas	2024	43.60465116	0.931	8.14	0	1	0	1	0	0	0	47	1	
153	8	Houston Dynamo	2024	39.72868217	0.767	7.38	0	1	0	1	0	0	0	55	1	
154	9	Inter Miami CF CF	2024	175.3875969	0.974	6.39	1	1	1	0	0	0	0	0	0	
155	10	LA Galaxy	2024	118.2170543	0.959	13.22	0	1	0	1	0	0	0	52	1	
156	11	LAFC	2024	0	1	13.22	0	1	0	1	0	0	0	57	1	
157	12	Minnesota United	2024	58.13953488	1	3.76	0	1	0	1	0	0	0	49	1	
158	13	New England Revolution	2024	38.75968992	0.442	5.03	0	1	0	1	0	0	0	44	0	
159	14	New York City FC	2024	63.95348837	0.715	19.83	0	1	0	1	0	0	0	45	0	
160	15	New York Red Bulls	2024	61.04651163	0.703	19.83	0	1	0	1	0	0	0	53	1	
161	16	Orlando City SC	2024	53.29457364	0.868	2.79	0	1	0	1	0	0	0	54	1	
162	17	Philadelphia Union	2024	53.29457364	1.019	6.33	0	1	0	1	0	0	0	58	1	
163	18	Portland Timbers	2024	67.82945736	0.879	2.59	0	1	0	1	0	0	0	50	1	
164	19	Real Salt Lake	2024	51.35658915	1.004	1.31	0	1	0	1	0	0	0	49	1	
165	20	San Jose Earthquakes	2024	42.63565891	0.972	2.06	0	1	0	1	0	0	0	46	0	
166	21	Seattle Sounders	2024	87.20930233	0.447	4.08	0	1	0	1	0	0	0	56	1	
167	22	Sporting Kansas City	2024	62.01550388	1.009	2.26	0	1	0	1	0	0	0	48	1	
168	23	Toronto FC	2024	67.82945736	0.829	6.8	0	1	0	1	0	0	0	38	0	
169	24	Vancouver Whitecaps	2024	22.28682171	0.492	2.82	0	1	0	1	0	0	0	52	1	

Figure 1 Table of cleaned variable data across 2018 to 2024 for the 24 teams.

The setup regression equation for the first research question is

$$Y_{it} = \alpha_i + \lambda_t + \beta_1(InterMiami_i \times MessiEra_t) + \beta_2(NonMiami_i \times MessiEra_t) + \gamma'X_{it} + \varepsilon_{it}$$

Where α_i is fixed team traits, such as stadium quality, market rating on the team; λ_t is the fixed shocks every year, such as market CPI, expansion, schedule change; Y_{it} is the total team revenue; $InterMiami_i$ is 1 if the team is Inter Miami, otherwise is 0. $NonMiami_i$ is 0 if the team is Inter Miami, otherwise is 0. $MessiEra_t$ is 1 after July 2023, otherwise is 0.

$\beta_1(InterMiami_i \times MessiEra_t)$ is the change in outcome after July 2023. $\beta_2(NonMiami_i \times MessiEra_t)$ is the average lift for the other teams after July 2023. $\gamma'X_{it}$ is the control of points, playoff, market size, and capacity.

Fixed-effects (within) regression
Group variable: **team_id**

Number of obs = 144
Number of groups = 24

R-squared:
Within = 0.5449
Between = 0.0088
Overall = 0.3284

Obs per group:
min = 6
avg = 6.0
max = 6

corr(u_i, Xb) = -0.0922

F(5, 23) = .
Prob > F = .

(Std. err. adjusted for 24 clusters in **team_id**)

Revenue_real	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
InterMiami_Post	104.8348	4.312851	24.31	0.000	95.91303	113.7566
NonMiami_Post	21.35709	5.299423	4.03	0.001	10.3944	32.31978
Year						
2019	.0608866	3.555835	0.02	0.986	-7.294918	7.416691
2020	-16.19008	3.487555	-4.64	0.000	-23.40463	-8.97552
2021	10.07069	4.843059	2.08	0.049	.0520604	20.08932
2023	-12.8918	5.29339	-2.44	0.023	-23.84201	-1.941585
2024	0	(omitted)				
_cons	37.1398	2.374491	15.64	0.000	32.22779	42.05181
sigma_u	15.989849					
sigma_e	15.330359					
rho	.52104705	(fraction of variance due to u_i)				

Figure 2 Result of the basic regression model from Stata.

After running the basic regression model indicated above for a total of 144 observations, the real revenue is shown. From the row $\beta_{1(InterMiami_i \times MessiEra_t)} = 104.83$, we know that after the big star's arrival, the team's revenue jumped hugely high, with an increase of about 105 million dollars, compared to pre-Messi time. From the row of $\beta_{2(NonMiami_i \times MessiEra_t)} = 21.36$, we know that other teams also experienced a visible increase of revenue compared to the time before Messi's arrival. By computing the magnitude ratio, $104.8/21.4 = 4.90$, it tells us that Inter Miami got the most gains, but other teams in the market also absorb some effect from the big star. After accounting for the effects within the market and the fixed effects, it is found that the positive impact remained significant. The year of the COVID-19 pandemic (2020) saw a substantial negative change in level, which is in expectations. Overall, the results demonstrated a statistically significant direct effect as well as a spread-out effect among all the teams.

In order to ensure the Messi spillover effect estimates are valuable, I run the robustness test to show that the direction and magnitude of the key coefficient change are valuable under assumptions. I swap the real revenue for log revenue to show that the result is not driven by other factors such as prices and quantities. I removed the 2020 data to make sure the estimates are not affected by the shock from COVID-19.

Fixed-effects (within) regression		Number of obs	=	144		
Group variable: team_id		Number of groups	=	24		
R-squared:		Obs per group:				
Within	= 0.5449	min	=	6		
Between	= 0.0088	avg	=	6.0		
Overall	= 0.3284	max	=	6		
		F(5, 23)	=	.		
corr(u_i, Xb) = -0.0922		Prob > F	=	.		
(Std. err. adjusted for 24 clusters in team_id)						
Revenue_real	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
InterMiami_Post	104.8348	4.312851	24.31	0.000	95.91303	113.7566
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2024	0	(omitted)				
_cons	37.1398	2.374491	15.64	0.000	32.22779	42.05181
sigma_u	15.989849					
sigma_e	15.330359					
rho	.52104705	(fraction of variance due to u_i)				

Figure 3 Table of outcome after changing real revenue to log revenue.

By adding the controls, it is confirmed that robustness of the spillover effect stay unchanged on other teams. Whether we change the conditions keep the direct effect of Messi's arrival stay, which is consistent with the baseline.

1.3 Conclusion

The regression model in 1.2 shows a significant increase for Inter Miami of around 105 million dollars and a positive effect of the big star spillover among other teams, even without Messi. The spillover stays after I add performance controls. Miami still shows a large premium. The capacity limit explains why revenue jumps more than attendance. The numbers support a short-run focus on price and product mix. The numbers also support pre-selling demand for the larger venue. As predicted, the team's revenue sits above our baseline for a typical team from MLS that shares similar trends. To conclude the economic mechanisms behind, it is counted for the big star effect, which shifts the demand up before the performance is fully done, and with the capacity limits of the stadium, the adjustment shows slowly rather than a big jump because of the mixed effect. Messi's extended influence goes beyond just the football market, lifting revenue above the prediction from a single market size or points alone. In the short term, the trend is still strong, as the direct effect of this superstar will last its pricing power and further shift up the commercial demand in the next one to two sessions. Since the stadium is fixed, constraints are still there, so the volume might stay the same in the short term. However, if the specific superstar is injured or an accident happens, the potential streaming can drop, thus leading to a decrease in the team's revenue.

Reference

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